Project: Worked examples
Title: Column base plate (EN 1993-1-8)
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1. Scope and method

An unstiffened rectangular base plate under an I/H, RHS or CHS column on grouted concrete: bearing resistance, the force regime and holding-down bolt tension for a major-axis moment, the tension T-stub of the plate and the compression T-stub, shear by friction and anchor bolts, and the plate thickness that satisfies them together. The bolt tension and shear per anchor go to Hilti PROFIS for the anchorage, which is done outside this sheet.

WHAT IT CALCULATES	RELATION
Bearing (cl. 6.2.5)	$f_{cd} = \alpha_{cc} \cdot f_{ck} / \gamma_c$; $f_{jd} = \beta_j \cdot \alpha \cdot f_{cd}$, $\beta_j = 2/3$; $c = t_p \cdot \sqrt{[f_y / (3 \cdot f_{jd} \cdot \gamma_{M0})]}$
Axial (cl. 6.2.8.2)	$N_{j,Rd} = f_{jd} \cdot A_{eff}$, A_{eff} the bearing strips of width c round the column footprint (full rectangle when they overlap); $u_N = N_{c,Ed} / N_{j,Rd}$
Moment (cl. 6.2.8.3, Table 6.7)	z_C from the section ($(h - t_f)/2$, $(h - t)/2$, half-annulus centroid) or entered, x_b the bolt row from the axis, $z = z_C + x_b$; $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ and $T_{Ed} = C_{Ed} - N_{Ed}$ when both are ≥ 0; with $N_{Ed} > 0$ and $T_{Ed} < 0$ both sides are in compression, $C = N_{Ed}/2 \pm M_{Ed}/(2z_C)$; with $N_{Ed} < 0$ and $C_{Ed} < 0$ both rows are in tension, $T = -N_{Ed}/2 \pm M_{Ed}/(2x_b)$; $F_{t,Ed} = T_{Ed}/n_t$ at the more loaded ro…
Plate T-stub (cl. 6.2.6.11, 6.2.4, Table 6.2)	$M_{pl,Rd} = 0.25 \cdot l_{eff} \cdot t_p^2 \cdot f_y / \gamma_{M0}$; without prying $\min(2 \cdot M_{pl,Rd}/m, n_t \cdot F_{t,Rd})$; with prying the modes $4 \cdot M_{pl,Rd}/m$, $(2 \cdot M_{pl,Rd} + n \cdot \Sigma F_{t,Rd})/(m + n)$ with $n = \min(e_{min}, 1.25m)$, and $n_t \cdot F_{t,Rd}$; l_{eff} is the total for the n_t bolts
Compression T-stub (cl. 6.2.6.9, 6.2.6.10)	$C_{Rd} = f_{jd} \cdot A_{flange}$, checked when $M_{Ed} > 0$ and the resultant is in compression
Shear (cl. 6.2.2)	$F_{2,vb,Rd} = \alpha_{bc} \cdot f_{ub} \cdot A_s / \gamma_{M2}$, $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$; $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$; $V_{Rd} = C_{f,d} \cdot N_{c,Ed} + n_b \cdot F_{vb,Rd}$; $V_{b,Ed} = \max(V_{Ed} - C_{f,d} \cdot N_{c,Ed}, 0)/n_b$
Required thickness	the smallest t_p in 10–100 mm with u_N, u_C and u_T all ≤ 1, z_C, C_{Ed} and T_{Ed} re-evaluated at every trial t_p

Basis: Design resistance of column bases in shear[1](cl. 6.2.2); Equivalent T-stub in tension[1](cl. 6.2.4, Table 6.2); Equivalent T-stub in compression[1](cl. 6.2.5); Concrete in compression including grout[1](cl. 6.2.6.9); Base plate in bending under compression[1](cl. 6.2.6.10); Base plate in bending under tension[1](cl. 6.2.6.11); Anchor bolts in tension[1](cl. 6.2.6.12); Column bases subject to axial forces[1](cl. 6.2.8.2); Column bases subject to axial force and bending moment[1](cl. 6.2.8.3, Table 6.7); Design strength of concrete; partial area loading[2](cl. 3.1.6, 6.7).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
column	Column section type	I/H		drawing: PF-GA-001 rev C	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h	Column depth, along the plate length L (I/H and RHS)	260.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	50 … 1200 mm
b	Column width, along the plate width B (I/H and RHS)	256.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	50 … 1200 mm
t_f	Flange thickness (I/H)	17.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	3 … 100 mm
t_w	Web thickness (I/H)	10.3	mm	spec: BS EN 10365 UC 254×254×89, to be checked against the owner's copy	3 … 100 mm
B	Plate width	400	mm	drawing: PF-BP-001 rev B	100 … 1500 mm
L	Plate length	450	mm	drawing: PF-BP-001 rev B	100 … 1500 mm
t_p	Plate thickness	25	mm	drawing: PF-BP-001 rev B	10 … 100 mm
f_y	Plate yield strength	265	MPa	spec: BS EN 10025-2 S275, 16 mm < t ≤ 40 mm	235 … 460 MPa
f_{ck}	Concrete characteristic cylinder strength	30	MPa	spec: BS EN 206 C30/37 pad foundation, drawing PF-FN-001	20 … 50 MPa
$f_{ck,g}$	Grout characteristic strength (optional: checks the β_j = 2/3 condition)	60	MPa	spec: Non-shrink grout datasheet, 28-day strength (placeholder GR-DS-01)	5 … 100 MPa
t_g	Grout thickness (optional: checks the β_j = 2/3 condition)	30	mm	drawing: PF-BP-001 rev B	1 … 200 mm
α_{cc}	α_{cc}, long-term effects on the concrete strength	0.85		spec: UK NA to BS EN 1992-1-1, NA to cl. 3.1.6(1)P	0.8 … 1
γ_c	γ_c, partial factor for concrete	1.5		spec: UK NA to BS EN 1992-1-1, Table NA.1	1.2 … 1.8

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
γ_{M0}	γ_{M0}, partial factor for the plate	1		spec: UK NA to BS EN 1993-1-1, NA.2.15	1 … 1.25
γ_{M2}	γ_{M2}, partial factor for the anchor bolts	1.25		spec: UK NA to BS EN 1993-1-8, NA.2.3	1.1 … 1.5
C_{fd}	$C_{f,d}$, friction coefficient plate to grout	0.2		spec: EN 1993-1-8 cl. 6.2.2, sand-cement grout	0 … 0.45
α	α, concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known)	1.5		assumption: Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_c1/A_c0 well above 2.25; checked on drawing PF-FN-001.	1 … 3
α_{basis}	Foundation geometry that α rests on (state it when α > 1)	Pad 1.2 × 1.2 m, 0.8 m deep (PF-FN-001): A_c1/A_c0 above 2.25, so α = 1.5 is taken		drawing: PF-FN-001 rev A	any
N_{Ed}	Axial force, COMPRESSION positive (the opposite sign to S3's N_{Ed}, which is tension positive; S2's $R1_y$, an upward reaction, is compression positive)	210	kN	reference: S5 load combinations sheet, ULS 6.10b, wind leading, column base A	-5000 … 10000 kN
M_{Ed}	Major-axis moment (uniaxial)	95	kN·m	reference: S5 load combinations sheet / S2 frame sheet, column base A	0 … 2000 kN·m
V_{Ed}	Shear force	60	kN	reference: S5 load combinations sheet / S2 frame sheet, column base A	0 … 2000 kN
n_t	Holding-down bolts in each bolt row (one row each side)	2		drawing: PF-BP-001 rev B	1 … 20
n_b	Holding-down bolts in total (both sides)	4		drawing: PF-BP-001 rev B	2 … 40

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
x_b	Bolt row distance from the column axis (outside the column face, inside the plate edge)	185	mm	drawing: PF-BP-001 rev B	50 … 1000 mm
m	Bolt to flange-weld toe distance, tension T-stub (required: it follows from x_b, the column and the weld)	46.9	mm	assumption: Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.	5 … 300 mm
l_{eff}	Effective length of the tension T-stub, total for the n_t bolts (required: from the cl. 6.2.6.5 patterns for the plate)	350	mm	assumption: Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).	10 … 2000 mm
prying	Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate	no		assumption: Cast-in holding-down bolts: prying not assumed (min(2·M_pl/m, ΣF_t) is the conservative case).	any
$F_{t,Rd}$	Tension resistance per bolt (from S4)	203	kN	reference: S4 bolt sheet, M24 8.8 (203.04 kN), rounded down	1 … 2000 kN
f_{ub}	Anchor bolt ultimate strength	800	MPa	spec: EN ISO 898-1 class 8.8	400 … 1000 MPa
f_{yb}	Anchor bolt yield strength (α_{bc} is valid for 235–640 MPa)	640	MPa	spec: EN ISO 898-1 class 8.8	235 … 640 MPa
A_s	Anchor bolt tensile stress area (from S4)	352.5	mm²	reference: S4 bolt sheet, M24 stress area (352.50 mm²)	50 … 1500 mm²
$F_{1,vb,Rd}$	Bolt bearing resistance on the plate, $F_{1,vb,Rd}$ (from S4 $F_{b,Rd}$)	364.4	kN	reference: S4 bolt sheet, M24 bearing on 25 mm S275 plate, e_1 = 60 mm, d_o = 27 mm (364.44 kN)	1 … 2000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$profis_{ref}$	Hilti PROFIS report reference (the report is filed as Evidence)	PROFIS-BP1 (to be filed)		assumption: The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.	any

3. Assumptions

A1. N_{Ed} is positive in compression here, the opposite of S3 (tension positive): a link from S2 must take the reaction $R1_y$ (up positive, compression into the base), never the member force N_{e1}. (Sign convention.)
A2. Static loading; a major-axis moment only (uniaxial); an unstiffened rectangular plate on grout of adequate strength and thickness for β_j = 2/3 (checked when the grout strength and thickness are entered). (Scope of S6.)
A3. The simplified T-stub model: one compression resultant at the centre of the compression T-stub (flange centre, or the half-annulus centroid of a CHS) and one tension resultant per bolt row; the force regime follows EN 1993-1-8 Table 6.7. The web is not given credit in the moment check. (EN 1993-1-8 cl. 6.2.8.3.)
A4. The effective length l_{eff} is the total for the n_t bolts of a row, read from the cl. 6.2.6.5 patterns; it is used for modes 1 and 2 alike ($l_{eff,1} \le l_{eff,2}$, so this is conservative for mode 2). (One input.)
A5. Concrete, anchor tension, pull-out, breakout, edge and the tension–shear interaction of the anchors are designed in Hilti PROFIS from $F_{t,Ed}$ and $V_{b,Ed}$ per anchor; the report is filed as Evidence and referenced. With prying switched on the prying force Q is not in $F_{t,Ed}$ and PROFIS must add it. (External anchorage design.)
A6. The friction resistance counts only the compression, $C_{f,d}\cdot\max(N_{Ed}, 0)$; a column in net tension has none. Anchor bolts share shear equally, with their holes assumed no larger than the standard clearance (oversized or slotted holes are outside this sheet). (EN 1993-1-8 cl. 6.2.2.)
A7. The bearing outstand c is cut back where it would pass the plate edge; the web and the column interior keep the full c. The CHS compression side is half of the annular bearing area, with its centroid as lever arm (an approximation). (Conservative against an uncut c.)
A8. The compression T-stub is checked against the flange area alone as soon as $M_{Ed} > 0$, while at $M_{Ed} = 0$ the whole footprint (web included) resists, so the compression check jumps at $M_{Ed} = 0^+$ (conservative). (Limitation.)
A9. Partial factors and the friction coefficient come from the records in use (γ_c, α_{cc}: EN 1992-1-1 NA; γ_{M2}: EN 1993-1-8 NA; γ_{M0}: EN 1993-1-1 NA; $C_{f,d}$: the grout used); the values shown default to the EN recommended and UK values. (A small set of factors only; no table from a standard is shipped.)
A10. α (α, concentration factor $\sqrt{(A_{c1}/A_{c0})}$ of the foundation (1 when the foundation geometry is not known)) is taken as 1.5. (Conservative: a 1.2 m square pad under the 0.45 × 0.40 m plate gives A_{c1}/A_{c0} well above 2.25; checked on drawing PF-FN-001.)
A11. m (Bolt to flange-weld toe distance, tension T-stub (required: it follows from x_b, the column and the weld)) is taken as 46.9 mm. (Bolt at 185 mm from the axis, flange face at 130.15 mm: 54.85 mm, less 0.8·√2 × 7 mm (a 10 mm fillet leg has a 7 mm throat) = 7.9 mm.)
A12. l_{eff} (Effective length of the tension T-stub, total for the n_t bolts (required: from the cl. 6.2.6.5 patterns for the plate)) is taken as 350 mm. (Total for the two bolts of the row, about 175 mm each from the cl. 6.2.6.5 patterns (engineer's estimate).)
A13. prying (Prying forces develop (short bolts, $L_b \le L_b^*$); off is conservative for the plate) is taken as no. (Cast-in holding-down bolts: prying not assumed ($\min(2\cdot M_{pl}/m, \Sigma F_t)$ is the conservative case).)
A14. $profis_{ref}$ (Hilti PROFIS report reference (the report is filed as Evidence)) is taken as PROFIS-BP1 (to be filed). (The PROFIS report is not attached in this example, so the sheet warns that the anchorage is not checked.)

4. Working

[31] $f_{cd} = \alpha_{cc} * f_{ck} / \gamma_c$ [2](cl. 3.1.6)
$= 0.85 \times 30\ \text{MPa} / 1.5$
$= 17\ \text{MPa}$
Concrete design compressive strength

[32] $\beta_j = 2/3$, fixed [1](cl. 6.2.5)
= grout conditions of cl. 6.2.5 assumed met
$= 0.66667$
Joint coefficient β_j

[33] $f_{jd} = \beta_j * \alpha * f_{cd}$ [1](cl. 6.2.5)
$= 0.66667 \times 1.5 \times 17\ \text{MPa}$
$= 17\ \text{MPa}$
Design bearing strength of the joint

[34] $c = t_p * \text{sqrt}(f_y / (3 * f_{jd} * \gamma_{M0}))$ [1](cl. 6.2.5)
$= 25\ \text{mm} \times \text{sqrt}(265\ \text{MPa} / (3 \times 17\ \text{MPa} \times 1))$
$= 56.987\ \text{mm}$
Additional bearing width round the column footprint

[35] $A_{eff} = 2(b + 2c)(t_f + 2c) + (h - 2t_f - 2c)(t_w + 2c)$ [1](cl. 6.2.5)
$= 2 \times 370.27 \times 131.27 + 111.73 \times 124.27\ \text{mm}^2$
$= 111100\ \text{mm}^2$
Effective bearing area

[38] $N_{j,Rd} = f_{jd} * A_{eff}$ [1](cl. 6.2.8.2, 6.2.5)
$= 17\ \text{MPa} \times 111100\ \text{mm}^2$
$= 1888.7\ \text{kN}$
Axial bearing resistance of the joint

[39] $N_{c,Ed} = \max(N_{Ed}, 0)$
$= N_{Ed} = 210\ \text{kN}$
$= 210\ \text{kN}$
Design compression in the column

[40] $u_N = N_{c,Ed} / N_{j,Rd}$
$= 210\ \text{kN} / 1888.7\ \text{kN}$
$= 0.11119$
Axial bearing utilisation

[41] $t_{req,N}$ = smallest t_p in 10–100 mm with $f_{jd} \cdot A_{eff}(c(t_p)) \geq N_{c,Ed}$ (bisection) [1](cl. 6.2.5)
$= N_{c,Ed} = 210\ \text{kN}$; carried at the 10 mm minimum
$= 10\ \text{mm}$
Plate thickness required for axial bearing alone

[42] $z_C = (h - t_f)/2$ [1](cl. 6.2.8.3)
$= (260.3 - 17.3)/2\ \text{mm}$
$= 121.5\ \text{mm}$
Compression resultant distance from the column axis

[43] $z = z_C + x_b$ [1](cl. 6.2.8.3)
$= 121.5 + 185$
$= 306.5\ \text{mm}$
Lever arm between the compression resultant and the bolt row

[44] $C_{Ed} = (M_{Ed} + N_{Ed} \cdot x_b)/z$ [1](cl. 6.2.8.3, Table 6.7)
$= (95\ \text{kN·m} + 210\ \text{kN} \times 185\ \text{mm})/306.5\ \text{mm}$
$= 436.7\ \text{kN}$
Compression resultant under the compression flange

[45] $T_{Ed} = C_{Ed} - N_{Ed}$ [1](cl. 6.2.8.3, Table 6.7)
$= 436.7 - 210$
$= 226.7\ \text{kN}$
Tension resultant at the bolt row

[46] $C_{Ed,2}$ = none
= other side
= 0 kN
Compression resultant under the other flange
[47] $T_{Ed,2}$ = none
= other row
= 0 kN
Tension resultant at the other bolt row
[48] $F_{t,Ed} = T_{Ed} / n_t$ [1](cl. 6.2.6.12)
= 226.7 kN / 2
= 113.35 kN
Holding-down bolt tension per bolt at the more loaded row (→ PROFIS)
[49] A_{flange} = (b + 2c)(t_f + 2c), at most $A_{eff}/2$ [1](cl. 6.2.6.9)
= min(370.27 × 131.27, $A_{eff}/2$) mm²
= 48608 mm²
Bearing area of the compression flange
[50] $C_{Rd} = f_{jd} * A_{flange}$ [1](cl. 6.2.6.9, 6.2.6.10)
= 17 MPa × 48608 mm²
= 826.33 kN
Bearing resistance of the compression flange
[51] $u_C = C_{Ed} / C_{Rd}$
= 436.7 kN / 826.33 kN
= 0.52849
Compression flange utilisation
[52] $M_{pl,Rd} = 0.25 * l_{eff} * t_p$^2 * f_y / γ_{M0} [1](Table 6.2)
= 0.25 × 350 mm × (25 mm)^2 × 265 MPa / 1
= 14.492 kN·m
Plastic moment resistance of the plate over l_{eff}
[53] $T_{Rd,1} = 2 * M_{pl,Rd} / m$ [1](Table 6.2)
= 2 × 14.492 kN·m / 46.9 mm
= 618 kN
Mode 1-2 without prying forces
[54] $T_{Rd,3} = n_t * F_{t,Rd}$ [1](Table 6.2)
= 2 × 203 kN
= 406 kN
Mode 3: bolt failure
[55] $T_{Rd} = \min(T_{Rd,1}, T_{Rd,3})$ [1](cl. 6.2.4)
= min(618 kN, 406 kN)
= 406 kN
Tension T-stub resistance (no prying)
[56] $u_T = T_{Ed} / T_{Rd}$
= 226.7 kN / 406 kN
= 0.55839
Plate bending and bolts in tension, T-stub utilisation
[57] $u_{bt} = F_{t,Ed} / F_{t,Rd}$
= 113.35 kN / 203 kN
= 0.55839
Holding-down bolt tension utilisation
[58] t_{req} = smallest t_p in 10–100 mm with u_N, u_C, $u_T \le 1$, with z_C, C_{Ed} and T_{Ed} recomputed at every trial t_p (scan and bisection) [1](cl. 6.2.5, 6.2.6.9, 6.2.6.11)
= $N_{c,Ed}$ = 210 kN, C_{Ed} = 436.7 kN, T_{Ed} = 226.7 kN
= 15.142 mm
Plate thickness required

[59] $\alpha_{bc} = 0.44 - 0.0003 \cdot f_{yb}$ [1](cl. 6.2.2)
$= 0.44 - 0.0003 \times 640$
$= 0.248$
Anchor bolt shear factor α_{bc}
[60] $F_{2,vb,Rd} = \alpha_{bc} * f_{ub} * A_s / \gamma_{M2}$ [1](cl. 6.2.2)
$= 0.248 \times 800\ MPa \times 352.5\ mm^2 / 1.25$
$= 55.949\ kN$
Anchor bolt shear resistance $F_{2,vb,Rd}$
[61] $F_{vb,Rd} = \min(F_{1,vb,Rd}, F_{2,vb,Rd})$ [1](cl. 6.2.2)
$= \min(364.4\ kN, 55.949\ kN)$
$= 55.949\ kN$
Anchor bolt shear resistance per bolt (bearing or shear)
[62] $F_{f,Rd} = C_{fd} * N_{c,Ed}$ [1](cl. 6.2.2)
$= 0.2 \times 210\ kN$
$= 42\ kN$
Friction resistance, compression only
[63] $V_{Rd} = F_{f,Rd} + n_b * F_{vb,Rd}$ [1](cl. 6.2.2)
$= 42\ kN + 4 \times 55.949\ kN$
$= 265.8\ kN$
Shear resistance of the base
[64] $u_V = V_{Ed} / V_{Rd}$
$= 60\ kN / 265.8\ kN$
$= 0.22574$
Shear utilisation
[65] $V_{b,Ed} = \max(V_{Ed} - F_{f,Rd}, 0)/n_b$ [1](cl. 6.2.2)
$= \max(60 - 42, 0)/4$
$= 4.5\ kN$
Shear per anchor bolt after friction (→ PROFIS)
[66] $u_{max} = \max(u_N, u_C, u_T, u_{bt}, u_V)$
$= u_N = 0.11119,\ u_C = 0.52849,\ u_T = 0.55839,\ u_{bt} = 0.55839,\ u_V = 0.22574$
$= 0.55839$
Governing utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
f_{cd}	17 MPa	Concrete design strength
β_j	0.66667	Joint coefficient β_j
f_{jd}	17 MPa	Bearing strength f_{jd}
c	56.987 mm	Bearing outstand c
A_{eff}	111100 mm²	Effective bearing area
$N_{j,Rd}$	1888.7 kN	Axial bearing resistance
u_N	0.11119	Axial bearing utilisation
$t_{req,N}$	10 mm	Plate thickness required for axial bearing alone
t_{req}	15.142 mm	Plate thickness required (bearing, compression and tension T-stubs)
z_C	121.5 mm	Compression lever arm z_C
z	306.5 mm	Lever arm z
C_{Ed}	436.7 kN	Compression resultant

SYMBOL	VALUE	DESCRIPTION
$C_{Ed,2}$	0 kN	Compression resultant, other side
T_{Ed}	226.7 kN	Tension resultant, more loaded row
$T_{Ed,2}$	0 kN	Tension resultant, other row
$F_{t,Ed}$	113.35 kN	Bolt tension per bolt (to PROFIS)
A_{flange}	48608 mm²	Compression-side bearing area
C_{Rd}	826.33 kN	Compression-side resistance
u_C	0.52849	Compression-side utilisation
$M_{pl,Rd}$	14.492 kN·m	Plate plastic moment
$T_{Rd,1}$	618 kN	T-stub mode 1-2, no prying
$T_{Rd,3}$	406 kN	T-stub mode 3
T_{Rd}	406 kN	Tension T-stub resistance
u_T	0.55839	Tension T-stub utilisation
u_{bt}	0.55839	Bolt tension utilisation
α_{bc}	0.248	Anchor shear factor α_{bc}
$F_{2,vb,Rd}$	55.949 kN	Anchor shear resistance $F_{2,vb,Rd}$
$F_{vb,Rd}$	55.949 kN	Anchor shear resistance per bolt
$F_{f,Rd}$	42 kN	Friction resistance
V_{Rd}	265.8 kN	Shear resistance
u_V	0.22574	Shear utilisation
$V_{b,Ed}$	4.5 kN	Shear per anchor bolt (to PROFIS)
u_{max}	0.55839	Governing utilisation
regime	"tension at one bolt row, compression under the other flange (Table 6.7, z = z_C + x_b)"	Force regime (Table 6.7)
T_{mode}	"mode 3 (bolt failure)"	Governing T-stub mode

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Base plate tension T-stub (bending at the bolts)	u_T = 0.55839	$u_T \le 1$	0.558	[1](cl. 6.2.6.11)
PASS	Holding-down bolt tension	u_{bt} = 0.55839	$u_{bt} \le 1$	0.558	[1](cl. 6.2.6.12)
PASS	Compression flange bearing	u_C = 0.52849	$u_C \le 1$	0.528	[1](cl. 6.2.6.9, 6.2.6.10)
PASS	Shear: friction plus anchor bolts	u_V = 0.22574	$u_V \le 1$	0.226	[1](cl. 6.2.2)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Concrete bearing under the axial force	u_N = 0.11119	$u_N \leq 1$	0.111	[1](cl. 6.2.8.2, 6.2.5)

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: α_{cc}, γ_c (EN 1992-1-1 National Annex factors); γ_{M2} (EN 1993-1-8 National Annex factor); γ_{M0} (EN 1993-1-1 National Annex factor); C_{fd} (Friction coefficient plate to grout); f_y (Plate strength). Enter each as one so its source, edition and licence travel with it.
WARNING — anchorage: $profis_{ref}$: 'PROFIS-BP1 (to be filed)' is typed text, not an attached report; attach the Hilti PROFIS report as Evidence so the reference points at the filed document. The anchorage design is not checked on this sheet.

8. Verification

Template struct.base-plate@1.2.0 Content hash 78225cb0777ee5bb Engine calc-engine/1.0.0
Known-answer cases passed 9/9 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.
2. EN 1992-1-1:2004.